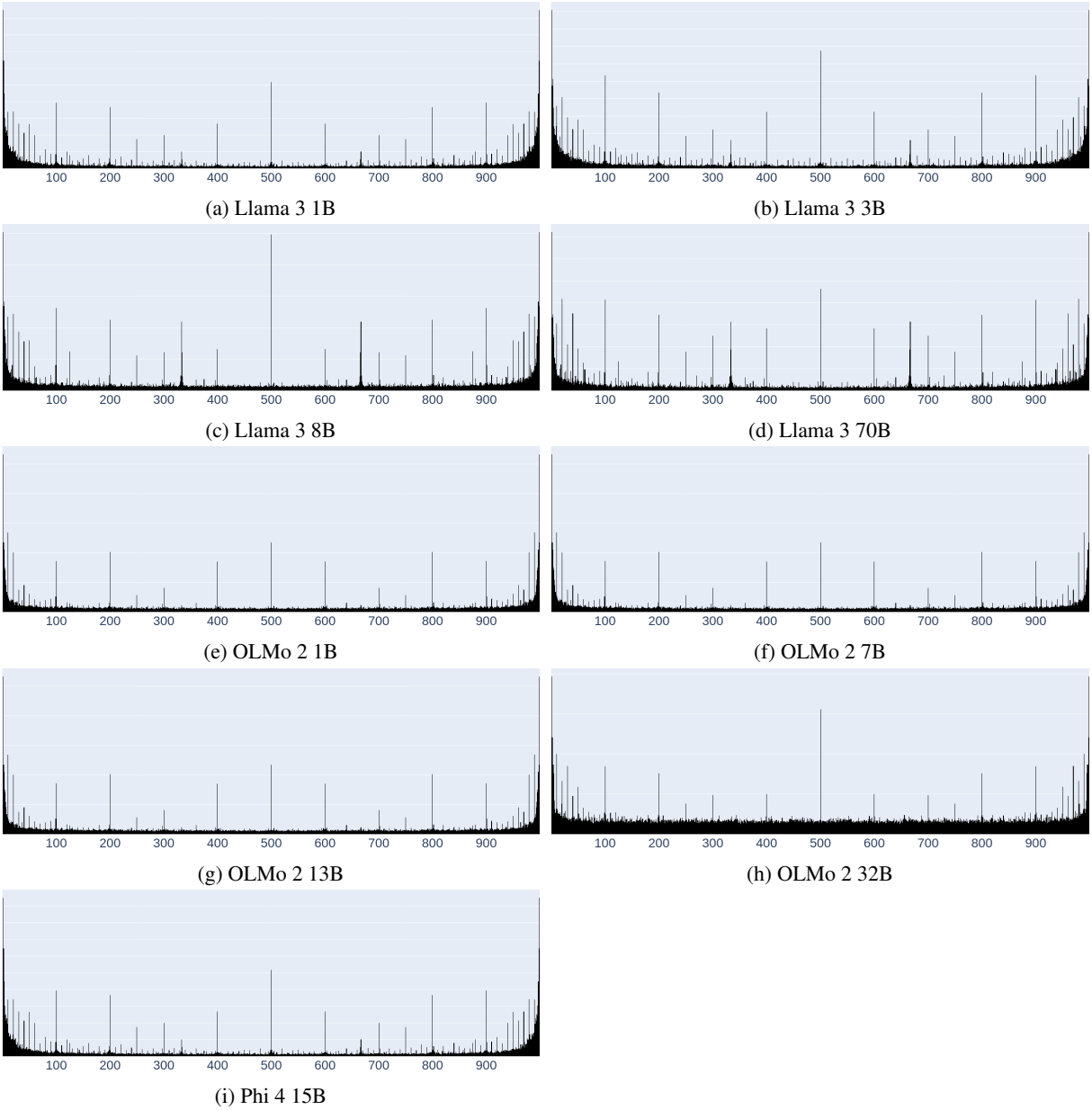
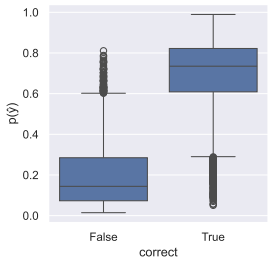
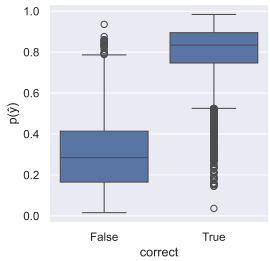



(a) Llama 3 1B (b) Llama 3 3B (c) Llama 3 8B (d) Llama 3 70B (e) OLMo 2 1B (f) OLMo 2 7B (g) OLMo 2 13B (h) OLMo 2 32B (i) Phi 4 15B

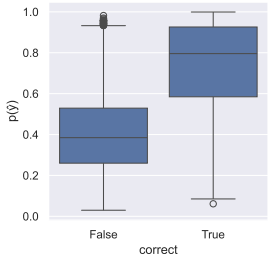
Figure 3: Maximal contribution (magnitude) of each Fourier base frequency's to embedding features in PCA (d=128) reduced space. Sparsity in this plot indicates strong regularity in the hidden structure of model embeddings. OLMo 2 32B has noticeably stronger contribution of all low-contribution frequencies, indicating high irregularity.



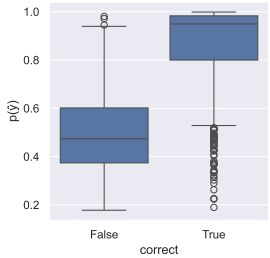

(a) Llama 3 1B, addition.




(b) Llama 3 1B, subtraction.

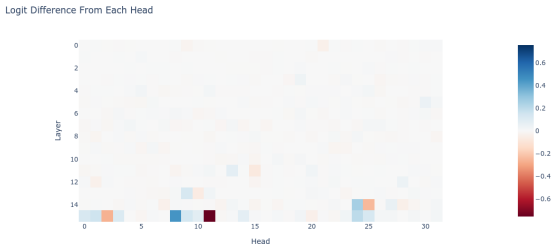



(c) OLMo 2 1B, addition.

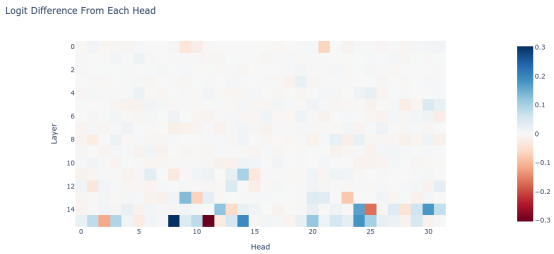



(d) OLMo 2 1B, subtraction.

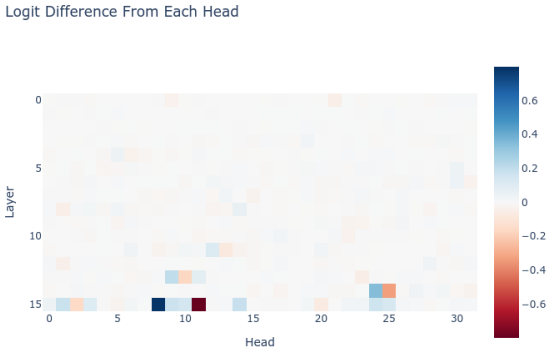
Figure 4: Probability mass on the predicted output token when the LLM yields a correct vs. incorrect answer.



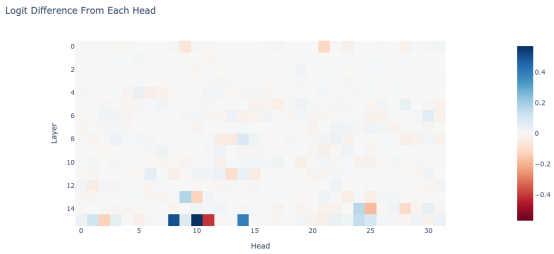

(a) Llama 3 1B, addition performed correctly.




(b) Llama 3 1B, addition performed incorrectly.

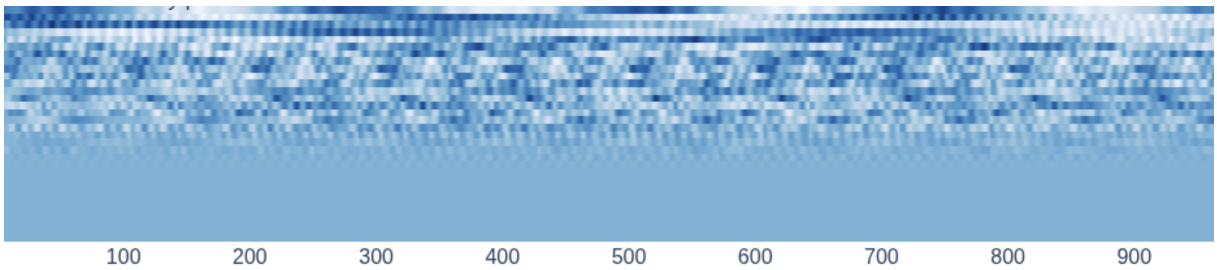



(c) Llama 3 1B, subtraction performed correctly.

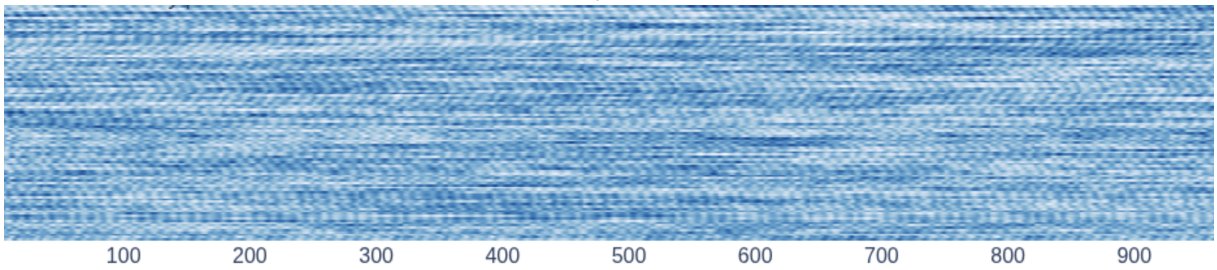



(d) Llama 3 1B, subtraction performed incorrectly.

Figure 5: Head activations across arithmetic tasks for Llama 3 1B, broken down by task (addition and subtraction) and success (correct or incorrect computation.




(a) L1 regularization




(b) L2 regularization

Figure 6: Hidden representations of sin-base probes for numeric input embeddings of Llama 3 1B model, after training with different regularization strategies show that our sin-base probes learn to project numeric embeddings into a generalized, wave-like representation used as target inductive bias.